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Agenda

- Chain of Thought (CoT)
- Program-aided Language Models (PAL)
- ReAct (Reasoning and Acting)
- LLM-powered Application Architecture

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Prompting

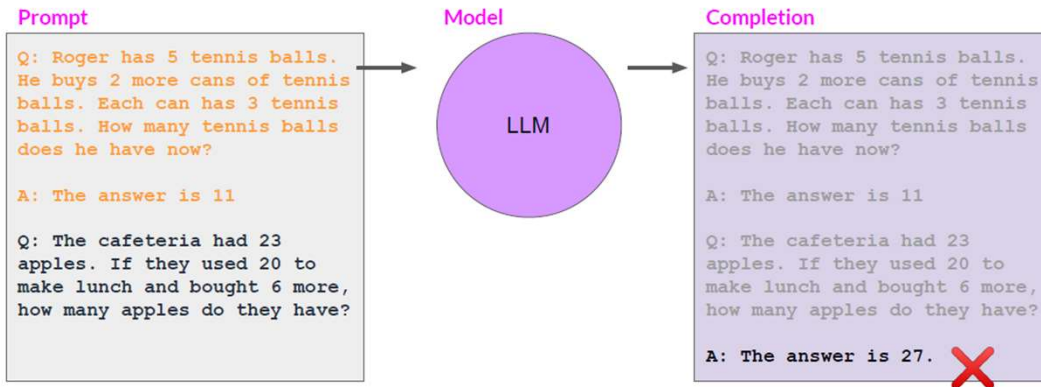
- Effective prompting is essential for maximizing the potential of large language models (LLMs).
- By crafting well-structured prompts, you can guide LLMs to perform complex tasks like question answering and reasoning with greater accuracy.
- It also helps in navigating their limitations, ensuring better results across a variety of applications.
- Developers and researchers rely on effective prompting to enhance the performance, safety, and functionality of LLMs, enabling them to integrate these models seamlessly with tools and domain-specific knowledge.

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Helping LLMs reason and plan with **Chain-of-Thought (CoT) Prompting**

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LLMs can struggle with complex reasoning problems



Slide Credits: DeepLearning.ai course: Generative AI and LLMs with AWS
<https://www.deeplearning.ai/courses/generative-ai-with-llms/>

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Humans take a step-by-step approach to solving complex problems

Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

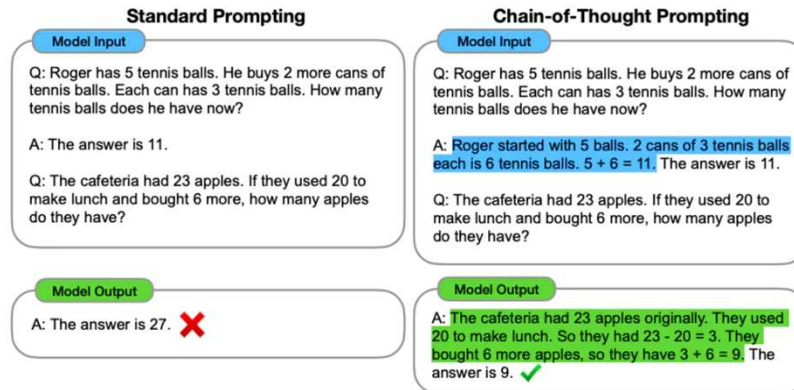
"Chain of thought"

Start: Roger started with 5 balls.
 Step 1: 2 cans of 3 tennis balls each is 6 tennis balls.
 Step 2: $5 + 6 = 11$
 End: The answer is 11

Reasoning steps

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Chain of Thought (CoT) Prompting



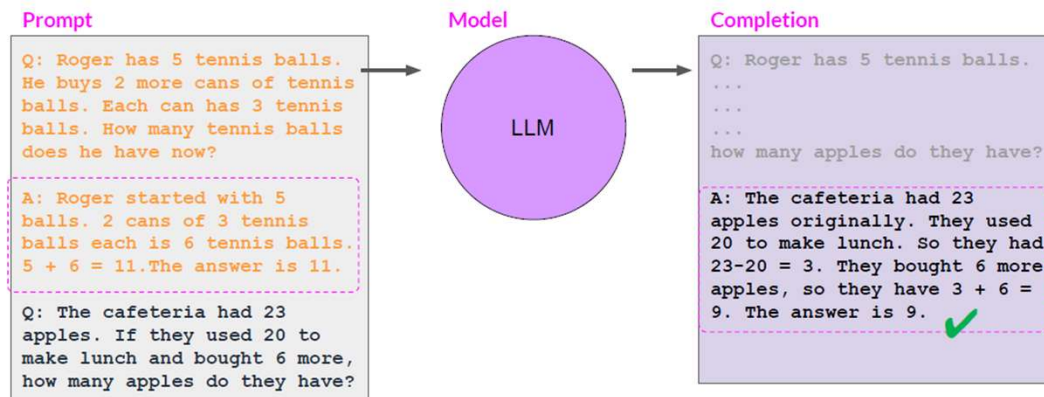
Introduced in [Wei et al. \(2022\)](#), chain-of-thought (CoT) prompting enables complex reasoning capabilities through intermediate reasoning steps.

You can combine it with few-shot prompting to get better results on more complex tasks that require reasoning before responding.

Source: Chain-of-Thought Prompting Elicits Reasoning in Large Language Models; Jason Wei et. al. NeurIPS 2022. [2201.11903 \(arxiv.org\)](#)

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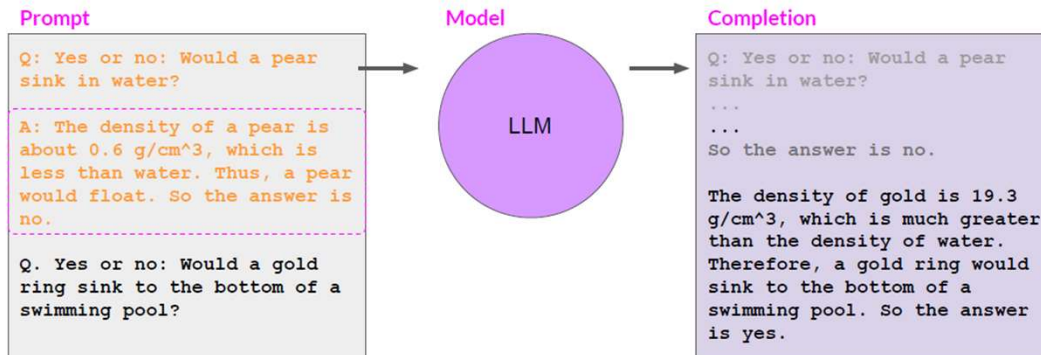
Chain-of-Thought Prompting can help LLMs reason



Source: Wei et al. 2022, "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models"

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Chain-of-Thought Prompting can help LLMs reason



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Program-aided Language (PAL) Models

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LLMs can struggle with mathematics

Prompt

What is $40366 / 439$?

Model

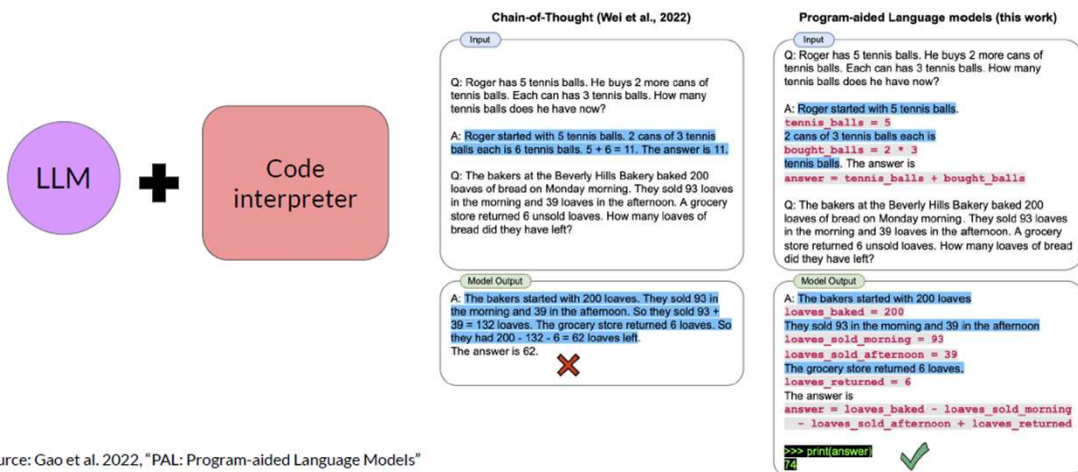
LLM

Completion

What is $40366 / 439$?
92.549

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Program-aided language (PAL) models



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PAL example

Prompt with one-shot example

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

Answer:

```
# Roger started with 5 tennis balls
tennis_balls = 5
# 2 cans of tennis balls each is
bought_balls = 2 * 3
# tennis balls. The answer is
answer = tennis_balls + bought_balls
```

Q: The bakers at the Beverly Hills Bakery baked 200 loaves of bread on Monday morning. They sold 93 loaves in the morning and 39 loaves in the afternoon. A grocery store returned 6 unsold loaves. How many loaves did they have left?

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PAL example

Prompt with one-shot example

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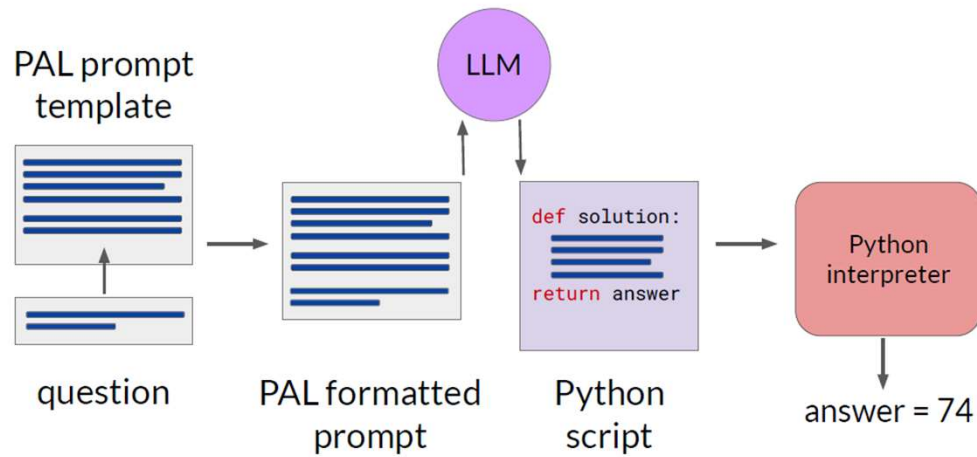
Completion, CoT reasoning (blue), and PAL execution (pink)

Answer:

```
# The bakers started with 200 loaves
loaves_baked = 200
# They sold 93 in the morning and 39 in the
afternoon
loaves_sold_morning = 93
loaves_sold_afternoon = 39
# The grocery store returned 6 loaves.
loaves_returned = 6
# The answer is
answer = loaves_baked
- loaves_sold_morning
- loaves_sold_afternoon
+ loaves_returned
```

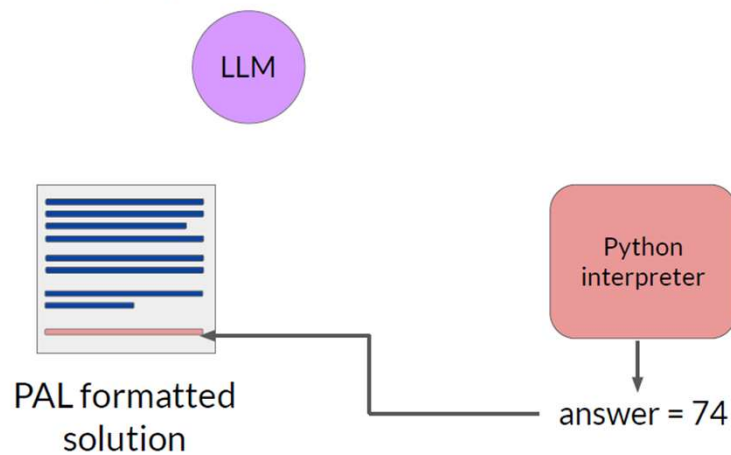
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Program-aided language (PAL) models



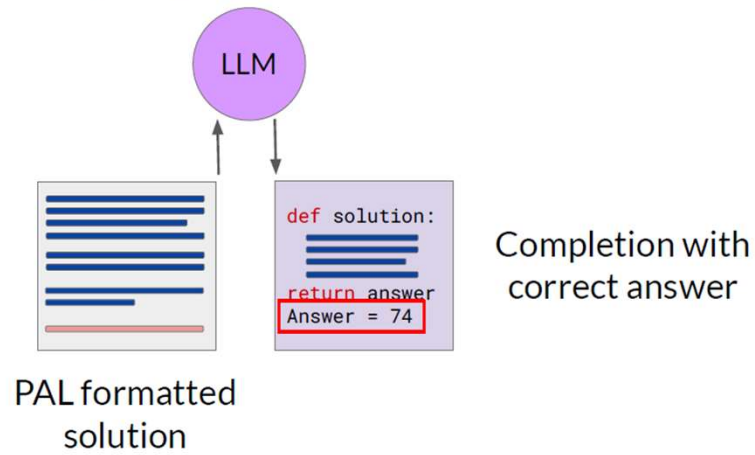
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Program-aided language (PAL) models



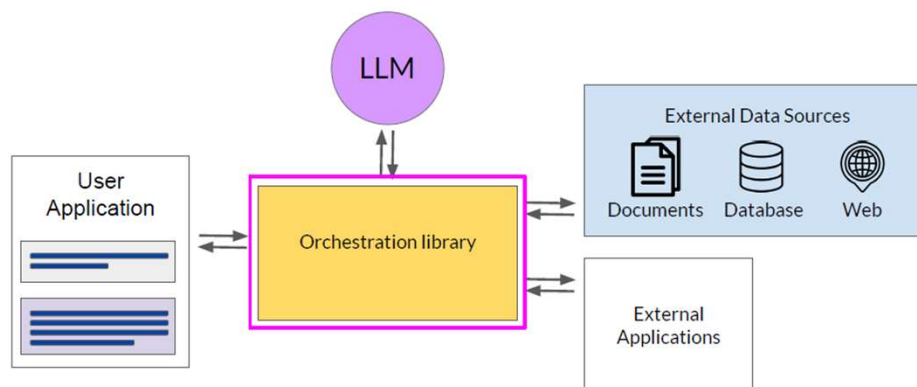
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Program-aided language (PAL) models



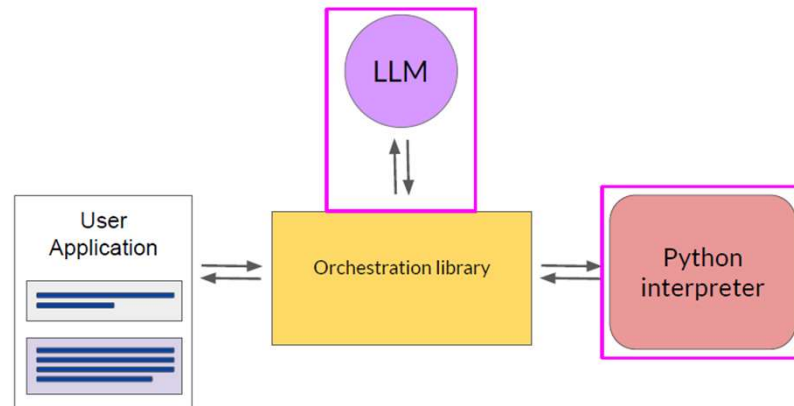
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LLM-powered applications



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PAL architecture



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ReAct: combining Reasoning and Action in LLMs

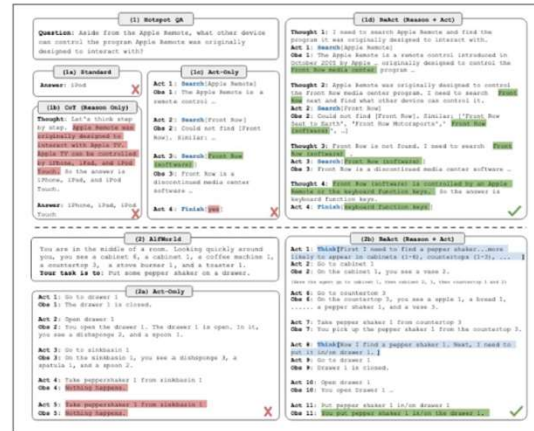
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ReAct: Synergizing Reasoning and Action in LLMs



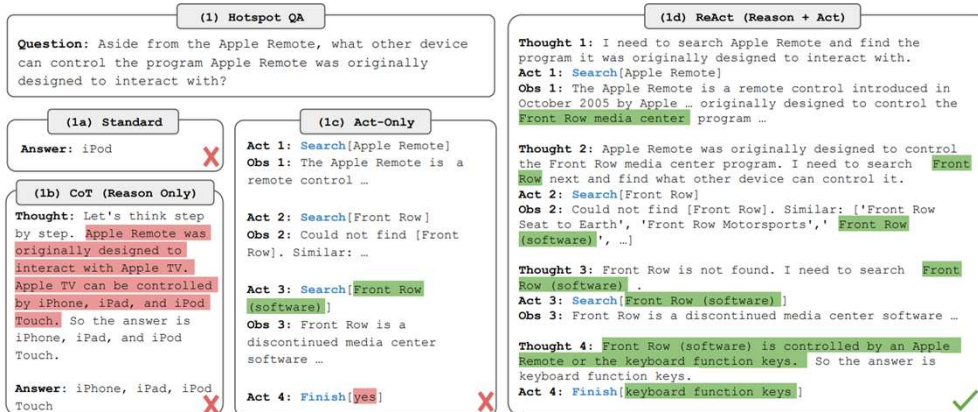
HotPot QA: multi-step question answering
Fever: Fact verification

Source: Yao et al. 2022, "ReAct: Synergizing Reasoning and Acting in Language Models"



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Reasoning and Acting (ReAct) Prompting



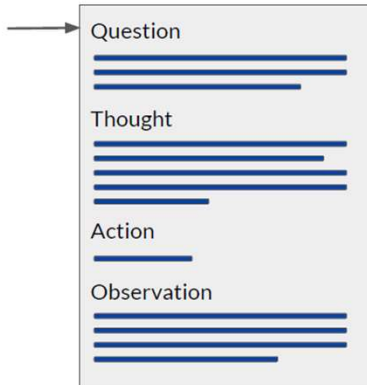
[Yao et al. 2022](#)

introduced a framework named ReAct where LLMs are used to generate both *reasoning traces* and *task-specific actions* in an interleaved manner.

Source: REACT: SYNERGIZING REASONING AND ACTING IN LANGUAGE MODELS; S Yao et. al. ICLR 2023.
[2210.03629 \(arxiv.org\)](#)

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ReAct: Synergizing Reasoning and Action in LLMs



Question: Problem that requires advanced reasoning and multiple steps to solve.

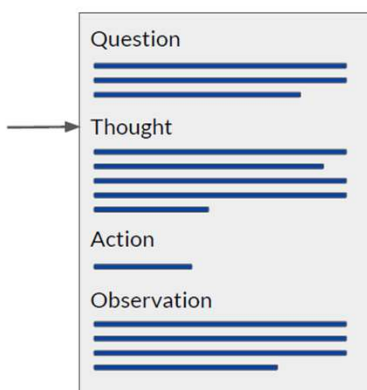
E.g.

"Which magazine was started first, *Arthur's Magazine* or *First for Women*?"

Source: Yao et al. 2022, "ReAct: Synergizing Reasoning and Acting in Language Models"

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ReAct: Synergizing Reasoning and Action in LLMs

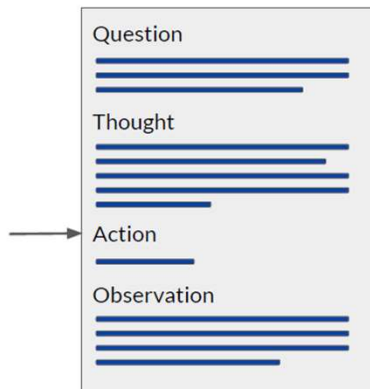


Thought: A reasoning step that identifies how the model will tackle the problem and identify an action to take.

"I need to search *Arthur's Magazine* and *First for Women*, and find which one was started first."

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ReAct: Synergizing Reasoning and Action in LLMs



Action: An external task that the model can carry out from an allowed set of actions.

E.g.

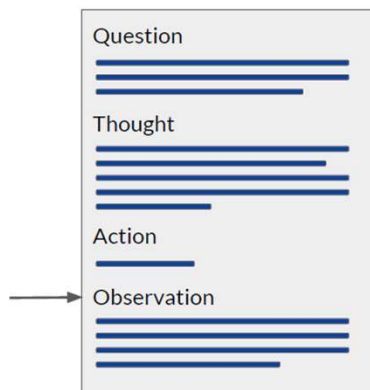
```
search[entity]
lookup[string]
finish[answer]
```

Which one to choose is determined by the information in the preceding thought.

```
search[Arthur 's Magazine]
```

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ReAct: Synergizing Reasoning and Action in LLMs



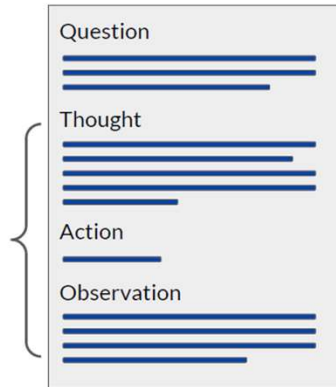
Observation: the result of carrying out the action

E.g.

"Arthur's Magazine (1844-1846) was an American literary periodical published in Philadelphia in the 19th century."

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ReAct: Synergizing Reasoning and Action in LLMs



Thought 2:

"Arthur's magazine was started in 1844. I need to search First for Women next."

Action 2:

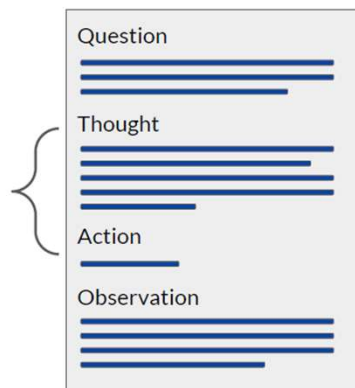
search[First for Women]

Observation 2:

"First for Women is a woman's magazine published by Bauer Media Group in the USA.[1] The magazine was started in 1989."

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ReAct: Synergizing Reasoning and Action in LLMs



Thought 3:

"First for Women was started in 1989. 1844 (Arthur's Magazine) < 1989 (First for Women), so Arthur's Magazine as started first"

Action 2:

finish[Arthur's Magazine]

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ReAct instructions define the action space

Solve a question answering task with interleaving Thought, Action, Observation steps.

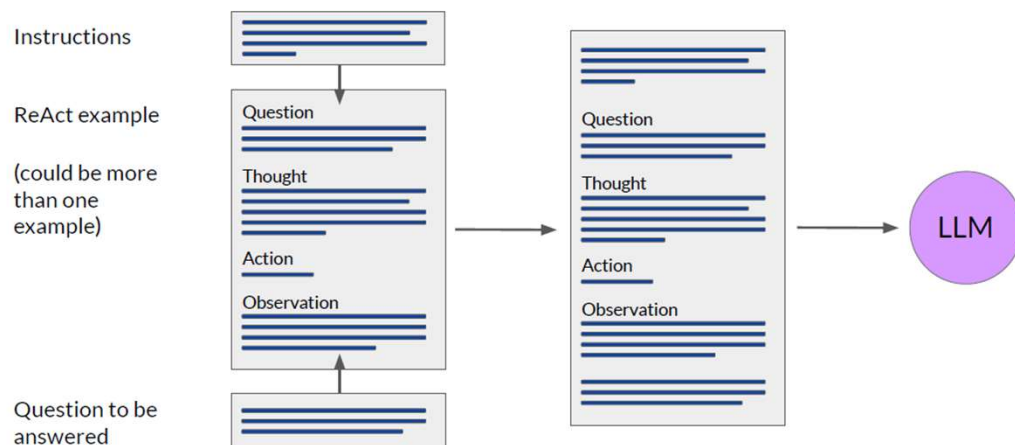
Thought can reason about the current situation, and Action can be three types:

- (1) `Search[entity]`, which searches the exact entity on Wikipedia and returns the first paragraph if it exists. If not, it will return some similar entities to search.
- (2) `Lookup[keyword]`, which returns the next sentence containing keyword in the current passage.
- (3) `Finish[answer]`, which returns the answer and finishes the task.

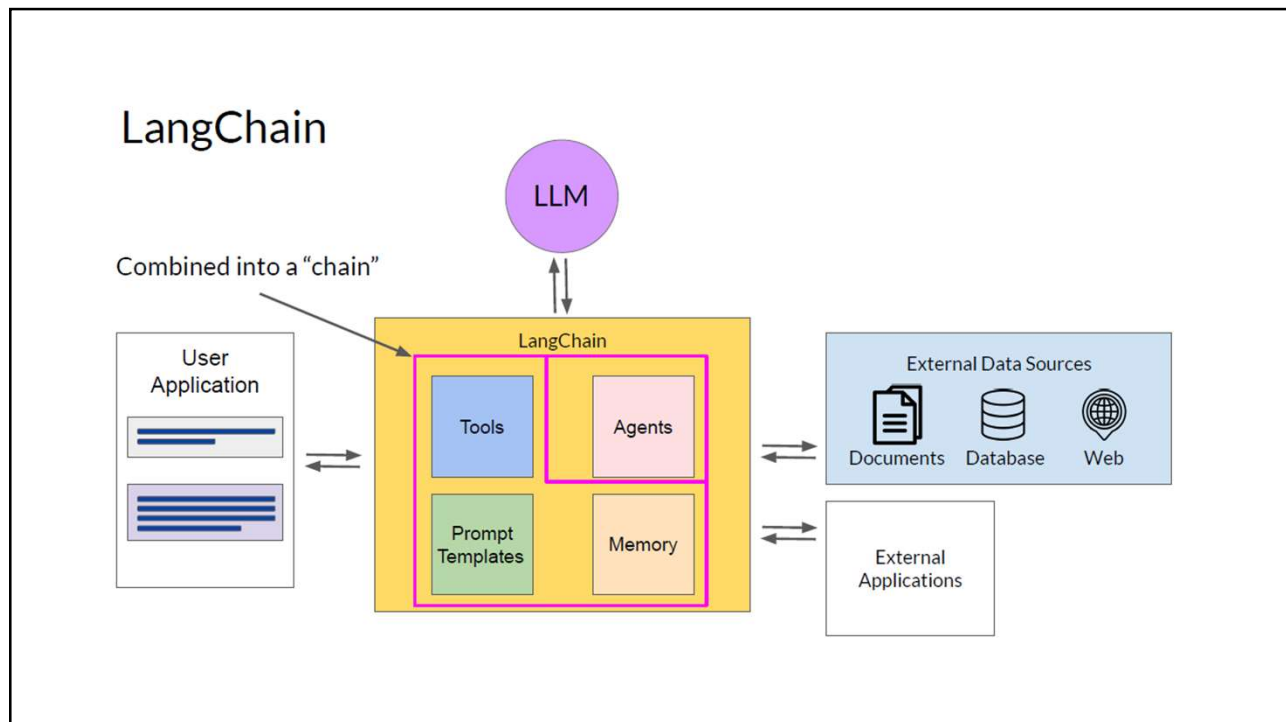
Here are some examples.

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Building up the ReAct prompt



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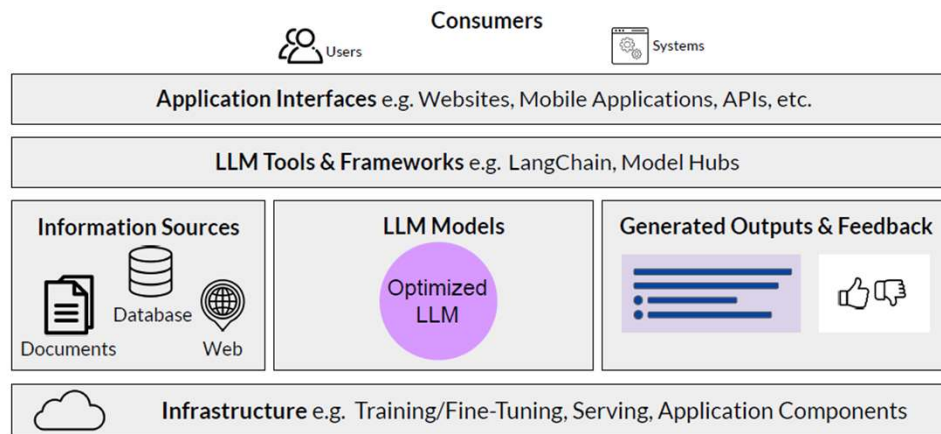


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LLM-powered Application Architecture

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Building generative applications



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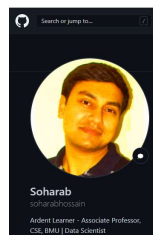
References

- **Slide Credits: DeepLearning.ai course: Generative AI and LLMs with AWS**
<https://www.deeplearning.ai/courses/generative-ai-with-llms/>
- **Prompt Engineering**
<https://lilianweng.github.io/posts/2023-03-15-prompt-engineering/>
- **ReAct Prompting**
<https://www.promptingguide.ai/techniques/react>
- **Chain of Thoughts (CoT)**
<https://www.youtube.com/watch?v=b210W3JWOxw>
- **LLM Reasoning**
<https://www.promptingguide.ai/research/llm-reasoning>

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Thank You

Class material will be uploaded here:
https://github.com/soharabhossain/GenAI_n_LLMs

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